The University of the South Pacific

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School of Computing, Information and Mathematical Sciences

CS211: Computer Organization

Mid-Semester Test – Semester 1, 2012

Time Allowed: 1½ hours

Total Marks: 55 (20% of final grade)

Name: _____ Student ID: _____

INSTRUCTIONS:

- There are 9 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

Question 1

a) What is the largest positive number that can be represented in 12 bits using one's complement? Represent the number in bits and decimal as well. Assume the first bit is the sign bit. (2 marks)

b) What is the smallest negative number that can be represented in 12 bits using one's complement? Represent the number in bits and decimal as well. Assume the first bit is the sign bit. (2 marks)

c) What is the smallest negative number that can be represented in 12 bits using two's complement? Assume the first bit is the sign bit. Represent the number in bits and decimal as well. (2 marks)

Question 2

Convert -15.5 to IEEE single-precision format. Represent the final answer in hexadecimal. (4 marks)

Question 3

For the truth table given below:

x	y	z	F(x,y,z)
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

a) Write the disjunctive normal form for the above truth table. (2 mark)

b) Draw the circuit diagram using the basic gates. (4 marks)

c) Implement the above using minimal number of gates. (4 marks)

Question 4

What would be the code word if you encode the data word 10110011 to allow one bit error detection and correction using hamming code with odd parity? (3 marks)

Question 5

A disk drive has 1024 cylinders, 10 tracks per cylinder, 32 sectors per track, and 512 bytes per sector. The rotation time is 4ms. The read/write head takes 10ms to move between adjacent cylinders and the maximum seek time is 300ms. A data file contains 800KB of data stored consecutively.
a) What is the minimum time needed to read the data? (3 marks)

b) What is the average time needed to read the data? (5 marks)

c) What is the maximum time needed to read the data? (3 marks)

Question 6

Show how AND function can be constructed from 3 NOR gates. (4 marks)

Question 7

What is the difference between Clocked SR-Latch and Clocked D-Latch? (2 marks)

Question 8

Write the output for the following machine language program if the inputs are 3, 2, 5 and -1. The inputs are in sequential order. Show the accumulator value for each input and any changes to it done by the program. (5 marks)

```
0011000000000000
1011000000001010
0110000000010000
0010000000010000
1001000000000000
0001000000010000
0100000000000000
0000000000000000
0000000000001010
```

Question 9 (10 marks)

Write a Wombat1 machine-language program that reads in one or more non-negative integers and outputs the average of the numbers (as an integer). It stops reading integers after it reads a negative integer. The negative integer is just a sentinel value and so is not to be included in the average. Your program should work with any number of non-negative integers. Do not write the program using binary numbers use decimal numbers instead. For example, for the machine instruction 0011000000000000, write "3 0" instead of 0011000000000000.

The End

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